DS605 · FUNDAMENTALS OF MACHINE LEARNING

Lecture Notes

Fundamentals of Predictive Analytics

Representation, Evaluation, and Optimization

Lecture 07–08

🛒 **Running example:** *Shopkart — the purchase-prediction model that will thread through the rest of the course*

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1. Where This Lecture Fits

Lectures	Theme
01–06	Data Mining + Mathematical Fundamentals
07 (this lecture)	What is supervised learning?
08 (this lecture)	How do we choose a hypothesis space?

Shopkart remains the linked example. We will keep returning to one business question so the mathematical symbols stay grounded.

2. Data Mining Tasks: Predictive vs. Descriptive

Descriptive patterns

Find human-interpretable structure in data.

- Examples: clustering, association patterns, sequence patterns
- Often related to unsupervised learning, but not always a one-to-one mapping

Predictive models

Use observed variables to predict unknown or future values.

- Examples: classification and regression
- Usually formalized as supervised learning

3. What Is Machine Learning?

Machine Learning is the study of computational methods that enable systems to acquire or improve models, representations, or behaviors through experience or data, rather than having them explicitly programmed.

4. Linked Example: Shopkart Purchase Prediction

Every Shopkart session can be described by three groups of features:

Group	Example attributes
Customer	age band, city, loyalty tier
Product	category, price band, discount
Session	search query, device, time, cart state

$$x = customer + product + session/context\ features$$

The purchase-prediction model's output *y* is binary: 0 = no purchase, 1 = purchase:

$$y \in \{0,\ 1\}$$

The model itself outputs a probability, not a hard label:

$$\hat{y} = P(purchase = 1 \mid x)$$

5. A Working Definition of Supervised Learning

At its simplest, training data teaches a model to map an unseen input to a prediction. Use this as a supervised-learning definition specifically — other ML settings, such as clustering and generative modeling, need broader definitions.

Programmed Rules vs. Learned Model

Conventional programming hard-codes the rules by hand. Supervised machine learning inverts this: rather than hard-coding every purchase rule, Shopkart learns a model from past sessions and their outcomes.

6. Training Examples: What Is One Data Point?

One Shopkart training example pairs an input with its observed outcome:

$$(x^{(i)},\ y^{(i)})$$

Symbol	What it holds
x(i)	Customer, product, session, and context features
y(i)	The observed label — purchase or no purchase, also called ground truth

Notation choice for the course: m = number of examples; d = number of features.

7. Supervised Learning: A Formal Model

Symbol	Meaning
X	Domain / instance space
Y	Label space
Γ : S → h	Learning algorithm — maps training data S to a hypothesis h
h : X → Y	Prediction rule / hypothesis

For a new product view *x*, predict $\hat{y} = h(x)$.

8. Deterministic vs. Stochastic Labels

Simplified / realizable view

$$x \sim D_X, \quad y = f(x)$$

Useful for first mathematical explanations.

General / noisy real-world view

Equivalently, either of two forms:

$$(x, y) \sim D$$

$$y \sim P(Y \mid X = x)$$

Better for Shopkart: Two similar customers may still behave differently.

Course convention: use y = f(x) when we need a simple target-function story; remember that real labels may be stochastic.

9. The IID Assumption and Stationarity

$$(X_i, Y_i) \sim D \ \ (iid), \quad i = 1, \ldots, m$$

Identically distributed

Each training example is generated by the same underlying distribution D.

Independent

Knowing one example should not change the probability law of the next example.

10. Learning = Representation + Evaluation + Optimization

This three-way decomposition will organize the rest of the course: every learning algorithm makes a choice about what it can represent, how it evaluates candidates, and how it searches for a good one.

Task	Representation	Evaluation	Optimization
Purchase prediction	Logistic regression	Log loss / AUC	Gradient descent
Purchase prediction	Decision tree	Validation accuracy / F1	Greedy split search
Order-value prediction	Linear regression	Mean squared error	Closed form or gradient descent
Product ranking	Scoring model	NDCG / revenue-aware metric	Gradient-based or heuristic search

11. Model Family, Hyperparameters, and Parameters

Model family / hypothesis space

- Decision trees
- Logistic models
- Neural networks

Hyperparameters filled in

Example: a decision tree with max_depth = 5.

Specific trained hypothesis

The actual learned splits, thresholds, and leaf predictions.

12. The Supervised Learning Pipeline

A training sample, filtered through a hypothesis space and guided by an evaluation objective, is handed to a learner that produces a final hypothesis, which then makes predictions.

The learner is not just “given data”; it is also constrained by a representation and guided by an evaluation objective.

13. Fit Is Not Enough: Generalization Is the Goal

Training fit

How well h performs on the examples used to learn it.

Generalization

How well h performs on new Shopkart sessions from the same process.

Evaluation

How we estimate generalization using validation/test data.

> **CONCEPT CHECK · IDENTIFY R, E, AND O**
>
> **Representation?** Decision-tree model class
>
> **Evaluation?** Validation F1-score
>
> **Optimization?** Greedy split search
>
> *Use this pattern whenever you meet a new ML algorithm.*

Bridge to Lecture 08 — a linear boundary, a shallow tree, a deep tree, and a neural network can all fit the same training data. Which one should we pick, and why? That question is exactly what a hypothesis space is for.

14. Choosing a Hypothesis Space: Why It's Hard

Recall the R–E–O lens from Lecture 07. Lecture 08 zooms into the "R": inductive bias, generalization, bias–variance, and complexity.

Shopkart Instance Space: How Big Is the Input World?

Attribute	Example values	Count
Customer tier	new, regular, loyal	3
Product category	electronics, fashion, home, beauty	4
Price band	low, medium, high	3
Discount level	none, low, high	3
Device	mobile, desktop	2
Time slot	morning, afternoon, evening, night	4
Cart state	empty, non-empty	2

Even a very small discrete version of Shopkart already creates many possible customer-product-session states (3×4×3×3×2×4×2 = 1,728). The real Shopkart feature space is much larger and often continuous.

Hypothesis Space Explosion

For binary classification, an unrestricted Boolean hypothesis assigns 0 or 1 to every instance in X:

$$\left|H_{all}\right| = 2^{|X|}$$

$$\left|H_{all}\right| = 2^{1728} \; for\ the\ small\ Shopkart\ example$$

That number already dwarfs anything that can be exhaustively searched. Three consequences follow directly:

- Impossible to search exhaustively
- Easy to overfit
- Needs inductive bias

15. Observed Data Alone Is Not Enough

Training data S

Covers only a tiny part of X.

Unseen instances

Still need predictions.

Any learner that generalizes beyond S must assume something about the structure of the world. That assumption is inductive bias, and the rest of this lecture is about where it comes from and how we choose it.

16. Hypothesis Space vs. Hypothesis

Candidate function families include: linear, polynomial, exponential, trigonometric, and decision tree.

The polynomial family, for instance, is an entire space of functions:

$$H_{poly} = \left\{\beta_0 + \beta_1 x + \cdots + \beta_k x^k\right\}$$

A single trained hypothesis is one specific member of that family:

$$h(x) = 2.1 - 0.7x + 3.4x^2$$

Model-Class Selection vs. Model Selection

	What it chooses	How
Model-class / representation selection	The family of functions to search within (e.g., logistic regression vs. decision tree vs. neural network)	Domain knowledge and engineering constraints
Model selection / hyperparameter selection	Among specific configurations (e.g., tree depth, regularization strength, architecture size)	Usually requires validation data

17. Two Ways to Induce Bias

Restriction bias

Limit the hypothesis space. Example: only allow linear decision boundaries.

Preference bias

Rank some hypotheses as more plausible than others. Example: prefer simpler models unless validation evidence says otherwise.

A Bayesian View: Preferences as Priors

Preference bias has a clean Bayesian reading — pick the hypothesis with the highest posterior probability given the data:

$$h* = argmax_{h \in H} P(h \mid S) \propto argmax_{h \in H} P(S \mid h) \cdot P(h)$$

Term	Meaning
P(S | h)	How well the hypothesis explains the observed data
P(h)	Prior preference for simpler, smoother, or domain-plausible hypotheses
P(h | S)	Posterior belief after seeing data

18. Choosing a Model Class: Subjective + Empirical Selection

Qualitative / problem-informed

- Business knowledge suggests interactions between customer, product, and context
- Interpretability may matter for merchandising teams
- Inference speed may matter for real-time recommendations

Quantitative / validation-based

- Compare candidate model classes and hyperparameters
- Choose tree depth or regularization strength using validation data
- Do not tune on the final test set

19. Experimental Evaluation: Fit vs. Generalization

Why not simply choose the most expressive model available?

Model	What happens
Too simple	Cannot capture important purchase patterns. High bias; underfitting risk.
Just right	Captures stable signals without memorizing noise. Best validation performance.
Too flexible	Fits accidental patterns in historical sessions. High variance; overfitting risk.

The learning objective: find a hypothesis that fits training data but, more importantly, generalizes to unseen data.

Training error

Error on data used to fit the model. Useful, but often optimistic.

Validation / test error

Error on held-out data. A better estimate of generalization — the next lecture formalizes evaluation protocols in full.

Formally, we distinguish two related quantities:

$$\hat{R}_S(h)$$

the *sample (empirical) error* — the average loss on the sample S.

$$R_D(h)$$

the *true (generalization) error* — the expected loss on new examples drawn from D.

Important qualification: *if h was fitted on S, the training error is usually too optimistic. To estimate generalization, use data not used to fit or select the model.*

20. The Bias–Variance Decomposition

The classical decomposition shown here is for squared-error regression at a fixed input x:

$$E\left[(Y - \hat{f}_S(x))^2\right] = Bias^2 + Variance + \sigma^2$$

Term	Meaning
Bias²	Systematic error due to assumptions/ restrictions
Variance	Sensitivity to the particular training sample
Irreducible noise σ²	Uncertainty in Y not removed by available X

Bias: Systematic Error from Simplifying Assumptions

$$Bias(x) = E_S\left[\hat{f}_S(x)\right] - f(x)$$

High bias — model is too restrictive; stable but systematically wrong. **Shopkart example:** a purely linear model may miss the nonlinear interaction between discount, product category, and customer tier. Often associated with underfitting, but bias itself is not always bad.

Variance: Sensitivity to Training-Sample Fluctuation

$$Variance(x) = E_S\left[(\hat{f}_S(x) - E_S\left[\hat{f}_S(x)\right])^2\right]$$

High variance — small changes in training sessions can produce a very different model. **Shopkart example:** an unrestricted decision tree can memorize accidental product-session quirks. Often associated with overfitting and poor performance on unseen data.

The Dartboard Analogy

Think of the target as true purchase behavior, and each dart as a prediction from a model trained on a different sample:

Regime	What it looks like
High bias, low variance	Consistently wrong
Low bias, high variance	Correct on average, but unstable
High bias, high variance	Wrong on average and unstable
Low bias, low variance	Accurate on average and stable — the goal

21. Underfitting and Overfitting in Shopkart

Underfitting

Model misses real purchase patterns. High bias, low flexibility.

Good fit

Model captures stable relationships and ignores noise.

Overfitting

Model memorizes accidental historical quirks. High variance, too flexible.

Signal in data	Too simple	Just right	Too flexible
Discount matters	misses it	uses it	overuses rare discount cases
Festival effects	misses seasonality	captures a stable seasonal pattern	memorizes one Diwali campaign
Rare products	ignores tail behavior	uses category-level signal	memorizes individual SKUs

22. Expressiveness, Complexity, and Sample Complexity

Expressiveness

The ability of H to represent many different functions or patterns.

Model complexity

How intricate the relationships are that a model can capture.

Capacity

Often used to refer to complexity and expressiveness together. For finite H, |H| is useful; parameter count is a heuristic, not a universal measure. (Advanced note: VC dimension is one formal capacity measure.)

Sample Complexity: Why Larger H Needs More Data

For a finite hypothesis class, the required sample size grows with the size of the hypothesis space and the desired confidence/accuracy:

$$m \geq \frac{1}{\varepsilon}\left[ln|H| + ln\frac{1}{\delta}\right]$$

This is the classic PAC guarantee — Probably (with probability ≥ 1 − δ) Approximately Correct (true error within ε). Exact constants and assumptions depend on the learning setting.

Quantity	What it controls
Hypothesis space size |H|	How many candidates the learner must tell apart — larger → more examples needed
Accuracy ε	Max tolerable gap between training error and true error — smaller → more examples needed
Confidence δ	Acceptable probability that the guarantee fails — smaller → more examples needed

Three Complexity Trade-offs

- **Statistical / sample complexity —** how much data is required to learn reliably?
- **Training complexity —** how expensive is it to find h from S?
- **Inference complexity —** how expensive is it to compute h(x) for a new user session?

Model	Training	Inference
Linear / logistic regression	moderate	fast
Decision tree	moderate	O(depth)
kNN	almost none	expensive
Deep neural network	expensive	bounded forward pass

Plotted against model complexity, training error keeps falling while validation error traces the classic U-shape — high on both ends, lowest at the sweet spot where bias and variance are balanced. The qualitative shape is shown here for intuition; it is the classic bias–variance curve, not fitted to measured data.

23. Learning as Search — Key Takeaways

Given a hypothesis space H, data S, and an inductive bias, learning becomes a guided search problem — the same R–E–O lens from Section 10, now fully specified:

Lens	Question
Representation	What can be learned?
Evaluation	What is good?
Optimization	How do we find it?

- Generalization requires inductive bias; no learner can infer unseen cases from S alone.
- Hypothesis-space choice is both domain-informed and empirical.
- Bias, variance, irreducible noise, and complexity explain underfitting and overfitting.
- Training error is not enough; evaluation requires held-out evidence.